

Executive Summary – Blinkit Sales & Outlet Performance Analysis

This project presents a comprehensive **SQL-based analysis of Blinkit's retail sales dataset**, focusing on **product performance, customer preferences, outlet characteristics, and sales distribution**. The objective of the analysis is to uncover **key business insights** that can help

Blinkit optimize **inventory planning, outlet strategy, product assortment, and revenue growth**.

The dataset was first **cleaned and standardized**, followed by multiple **aggregation and performance queries** to evaluate sales trends across **item fat content, product categories, outlet types, outlet sizes, locations, and establishment years**.

Overall, the analysis reveals that **product composition (fat content), outlet location, and outlet type** play a critical role in driving sales and customer ratings.

Project Objectives

1. Clean and standardize inconsistent product data
2. Measure overall sales, average sales, item count, and customer ratings
3. Analyze sales performance by:
 - Item Fat Content ○ Item Type ○
Outlet Location Type ○ Outlet Size ○ Outlet Type ○ Outlet Establishment Year
4. Identify high-performing and underperforming segments
5. Provide **data-driven recommendations** for business growth

Data Preparation & Cleaning

- Created a structured database **blinkitdb**
- Designed a normalized **blinkit table** with sales, product, and outlet attributes
- Standardized **ItemFatContent** values:
 - LF, low fat → **Low Fat** ○ reg → **Regular**
- Ensured consistent categorical data for accurate aggregation and reporting

This cleaning step improved **data reliability and analysis accuracy**.

Key Metrics Overview

Metric	Insight
Total Sales	Calculated in million format for executive readability
Average Sales per Item	Indicates typical product-level revenue
Total Items	Shows dataset size and product diversity
Average Rating	Reflects overall customer satisfaction

Detailed Analytical Findings

1. Fat Content Analysis

- Both **Low Fat** and **Regular** products contribute significantly to total sales
- Regular fat items generally show:
 - Higher total sales
 - average sales Slightly higher
- Low-fat items maintain **competitive ratings**, indicating strong demand among healthconscious consumers

Insight: Consumer demand is diversified — Blinkit benefits from maintaining both categories.

2. Top Item Types by Sales

- Top 5 item categories generate a **disproportionate share of revenue**
- These categories show:
 - High total sales
 - Strong average ratings
 - Large product count

Insight: A small set of product categories drives most revenue (Pareto effect).

3. Outlet Location Type Performance

- Sales vary significantly across location tiers
- Urban and semi-urban locations typically outperform rural ones
- Regular fat products dominate sales across all locations

Insight: Location strategy has a strong impact on revenue potential.

4. Outlet Establishment Year Analysis

- Older outlets tend to:
 - Generate higher total sales
 - Maintain stable customer ratings
- Newer outlets may show lower total sales but have growth potential

Insight: Outlet maturity positively impacts performance due to customer trust and brand familiarity.

5. Outlet Size Contribution

- Medium and large outlets contribute the **highest percentage of sales**
- Small outlets generate lower revenue share

Insight: Store capacity and assortment size directly influence sales volume.

6. Outlet Type Performance

- Certain outlet types dominate:
 - Total sales
 - Sales percentage
 - Average item visibility
- Outlet types with higher item visibility show better sales performance

Insight: Merchandising and shelf visibility strongly influence purchasing behavior.

Business Recommendations

1. Inventory Optimization

- Increase stock depth for **top-performing item types**
 - Maintain balanced inventory for **Low Fat and Regular** products to cater to diverse customers
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2. Outlet Strategy Enhancement

- Focus expansion on **high-performing outlet types and locations**
 - Upgrade small outlets in high-demand locations with better assortments
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3. Visibility & Merchandising

- Improve **item visibility** in underperforming outlets
 - Use data-driven shelf placement for high-margin products
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4. New Outlet Growth Strategy

- Provide promotional support to **newer outlets**
 - Use insights from older, high-performing outlets as benchmarks
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5. Customer Satisfaction Focus

- Monitor ratings alongside sales to:
 - Improve low-rated but high-selling items
 - Protect high-rated productswith consistent quality
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Conclusion

This SQL project demonstrates how **structured data analysis can transform raw sales data into meaningful business insights**. By combining **data cleaning, aggregation, and performance analysis**, the project highlights opportunities for:

- Revenue growth
- Operational efficiency
- Smarter outlet expansion
- Customer-centric product planning